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network can learn about itself and its topology, and use that to optimise the way it sends data, how far messages will go and so on.

Devices that are part of the provisioned network are called nodes. Regardless of the type of product, nodes can fulfil a variety of network functions, whether it is a light, light switch or thermostat. In relay nodes, messages get relayed (hopping through the network) only by nodes that are acting as relay nodes. This is a software feature, and not all devices or nodes in the network are relays. You can decide when you are designing and implementing your network, which devices will provide that service.

Friendship in Bluetooth network works like this. In any network, particularly a smart building, there are devices that are power-rich such as lights. These have limitless power because these are connected to the grid. And then there are low-power nodes, such as sensors, which are embedded on a surface and, hence, their batteries cannot be changed every time. Managing their power consumption is absolutely critical.

So, if a low-power node is equipped with a temperature sensor and is supposed to report temperature only when it goes above a certain limit or below, then it is a very efficient sensor because it is going to send one byte of data every three months, or something like that. Hence, there is no problem. But, if one of the requirements is that you must be able to change those thresholds by sending a message to the sensor whenever you want to, there will be an efficiency problem because the sensor needs to use the radio to listen for messages a lot of the time.

Friendship involves low-power nodes discovering devices that act like friends. It is a software feature that can send out messages to any other device, and these two devices work together. Messages that get sent to the low-power sensor actually get stored by the friend, which will accrue messages addressed to the sensor until the sensor wakes up and receives messages. So, devices can discover and agree to help each other.

The last network role a node can perform is called proxy, which is really

important because proxy lets any compatible Bluetooth Low Energy device in the market, say, a smartphone, and talk to a Bluetooth Mesh network without waiting for Google or Apple to upgrade their stack with new APIs. Standard APIs that are already available under those platforms can be used to talk to the rest of the network via a proxy node.

There are multiple proxy nodes all over the building, and standard Bluetooth Low Energy applications that work with the mesh network can be written. So, range of compatibility increases with every device. That is a major advantage of this technology.

Bluetooth Low Energy versus Bluetooth Mesh

In the world of Bluetooth Low Energy, security is optional. It is all about two devices, and the question about the type of security required is something only the manufacturers can decide. It is like a toolbox; there are lots of security capabilities, but manufacturers decide which of those capabilities they will use for their products.

Bluetooth Mesh is different because no manufacturer is allowed to reduce the security of the network. All messages are encrypted and authenticated. This means messages in the network cannot be spoofed, and so all communication is secure. It is possible to separate the security of different applications by different application keys, one for each mesh application. Devices only get application keys that are relevant to the application they are a member of.

Networks can be divided into sub-nets and different areas can be isolated cryptographically. This is great for hotels where each individual room has its own sub-net key issued at the time a booking is made. Messages are also obfuscated. This is a privacy mechanism, and network analysis attacks and figuring out patterns of communication are not possible.

The whole thing starts with the provisioning process, which is how devices get those keys in the first place.

Bluetooth Mesh is a major new wireless communication technology, which we are already seeing adopted in the emerging world of the smart building. **EFY**